ARUMUGAM **ABILASHAN**

instrumantation and automation

About Me

A highly motivated Instrumentation and Automation Engineering student seeking an internship to apply theoretical knowledge and gain practical experience. Proficient in robotics, SolidWorks, PCB design, and industrial automation. Eager to contribute to innovative projects and contribute to team success.

Contact

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Education

University of Colombo
Bachelor of Engineering Technology Honors in Instrumentation and Automation

CP/N/Cambridge collage kotagala.
General Certificate of Education Advance Level. Engineering technology

Major Skills

- Embedded systems
- Solidwork
- PCB designing
- FPGA programing
- MATLAB
- IOT
- SCADA

Selected Undergraduate Projects as a team

Atmega328P based embedded system

(Atmega328P-Based Temperature Monitoring System)

Core Functionalities:

- Real-time temperature sensing and display in Celsius and Fahrenheit on a 7-segment display.
- Audible alarm activation for temperatures exceeding 40°C.
- Visual temperature range indication using LEDs with varying brightness.
- Switchable modes for temperature display and range indication.

Development Tools:

- SimulIDE for virtual simulation.
- Physical implementation on a dot board

Arduino based mini project

Portable Luggage Safety Alert System)

- **Integrated Technologies:** Bluetooth, GSM, and GPS
- **Core Functionality:** real-time tracking and SMS alerts upon Bluetooth disconnection.
- **Hardware Components:** Arduino, HC-06, SIM900A, GY-GPS6MV2
- **Applications:** Personal and asset safety

Computer mini project -web applcation

hotel management System)

Core Functionalities:

- We've developed a user-friendly hotel management system to streamline operations and enhance guest experience.
- features include:

Efficient Reservations,Housekeeping Optimization,Financial Management,Data-Driven Insights,Streamlined Check-in/Checkout

Development Tools:

- Web development (HTML, CSS, JavaScript)
- Database management (SQL)
- Server-side scripting (PHP, Python, Node.js)

Language

Tamil 100%

English 70%

Sinhala 70%

Soft Skills

- Problem solving
- creative thinking
- communication
- team work
- sports

Work experience

Worked as an Electrician.
(Domestic Electricity wiring and maintenance (single and three phase.
Worked as a CCTV technician.

Selected Undergraduate projects—individual

DAQ System for Robotic Manipulator Monitoring

Objective:
Designed a DAQ system to monitor motor temperature, vibration, and acceleration in a robotic manipulator.
Tools:
Arduino, ADXL335 accelerometer, temperature sensor LM35 ,sw-420 vibration sensor, LabVIEW.
Features:

- Real-time data acquisition and visualization using LabVIEW.
- Integrated sensors for monitoring temperature, position, and force.
- Sensor calibration and error correction for accurate analysis.
- Dashboard for trend analysis and predictive maintenance.

solidwork- Quadruped robot design

- simulates a 4-legged walking robot. It demonstrates how the legs articulate and coordinate movement for forward/backward motion, turning, and potentially obstacle avoidance. This model is a valuable tool for visualizing and analyzing the mechanics of such robots.

solidwork- industrial robot arm design

- simulates an industrial robot arm, capable of multi-axis motion for precise positioning and manipulation of objects. This is achieved through joints like rotary or prismatic, allowing for complex movements. The design would incorporate considerations for load capacity, reach, speed, and accuracy.

Technical Skills and Tools

- **Electronics:** FPGA programming, Simul IDE, OrCAD Capture CIS,PCB design
- **Data Analysis & Statistics:** MATLAB, SPSS, Octave.
- **Engineering Design:** CAD, SolidWorks, fusion360 , Catia V5
- **Programming:** C, Python,C++
- **Web & Mobile Development:** HTML, CSS, JavaScript, Android/iOS development
- **Mathematics & Physics:** Calculus, Differential equations, Statistics, Applied Numerical methods, DSP, EMF and their applications
- **Robotics:** Sensors, Actuators, Kinematics, ROS
- **Embedded systems:** Atmel AVR, ESP32, Arduino

References

Dr. B.L.SANJAYA TILAKARATHNE
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